

Playwright XPath Locators — Notes

A practical guide to XPath locators in Playwright, based on the Demo Web Shop example.

1. XPath Locator

XPath is a locator strategy used to identify HTML elements. In Playwright, use `page.locator()` with an XPath expression.

```
const logo: Locator = page.locator("//img[@alt='Tricentis Demo Web Shop']");
```

2. Relative XPath

Relative XPath normally starts with `//` and searches for an element based on attributes or relationships. It is generally more maintainable than absolute XPath.

```
page.locator("//img[@alt='Tricentis Demo Web Shop']");
```

3. Absolute XPath

Absolute XPath starts from the root and normally begins with `/`. It depends heavily on the page structure and can break when the DOM changes.

```
page.locator("/html/body/div[4]/div[1]/div[1]/div[1]/a/img");
```

4. contains()

`contains()` is useful when an attribute has a dynamic or partially known value.

```
page.locator("//h2/a[contains(@href, 'computer')]");  
// Example: //*[contains(@class, 'btn')]
```

5. count()

`count()` returns the number of elements matched by a Locator.

```
const counter: number = await products.count();  
expect(counter).toBeGreaterThan(0);
```

6. first()

`first()` selects the first element from multiple matching elements.

```
await products.first().textContent();
```

7. nth()

`nth(index)` selects one element by index. Playwright indexing starts at 0, so `nth(0)` is the first element and `nth(3)` is the fourth.

```
await products.nth(3).textContent();
```

8. allTextContents()

`allTextContents()` returns the text content of all matching elements as an array.

```
const productTitles = await products.allTextContents();
```

9. Strict Mode

If a locator matches multiple elements, a single-element action can cause a strict mode violation. Narrow the locator with `first()`, `nth()`, or another condition.

```
await products.first().click();  
// or  
await products.nth(2).click();
```

10. Complete Example

```
import { test, expect, Locator } from "@playwright/test";

test("XPath Locators", async ({ page }) => {
  await page.goto("https://demowebshop.tricentis.com/");
  await page.waitForTimeout(3000);

  // Relative XPath
  const logo: Locator = page.locator("//img[@alt='Tricentis Demo Web Shop']");
  await expect(logo).toBeVisible();

  // Absolute XPath
  const abspath: Locator = page.locator("/html/body/div[4]/div[1]/div[1]/div[1]/a/img");
  await expect(abspath).toBeVisible();

  // XPath with contains()
  const products: Locator = page.locator("//h2/a[contains(@href, 'computer')]");
  const counter: number = await products.count();
  console.log("No of computer related products", counter);
  expect(counter).toBeGreaterThan(0);

  console.log("First computer related product", await products.first().textContent());
  console.log("Fourth computer related product", await products.nth(3).textContent());

  const productTitles = await products.allTextContents();
  console.log(productTitles);
});
```

11. Quick Revision Cheat Sheet

Method	Purpose
//tag[@attr='value']	Relative XPath using an exact attribute
/html/body/...	Absolute XPath from the document root
contains(@attr,'text')	Match part of an attribute value
count()	Count matched elements
first()	Select the first matched element
nth(0)	Select the first element
nth(3)	Select the fourth element
allTextContents()	Get text from all matched elements
textContent()	Read text from a selected element

Best practice: Prefer Playwright user-facing locators such as `getByRole()`, `getByLabel()`, `getByText()`, and `getByTestId()` when they provide a stable and meaningful selector. Use XPath when it is clear and maintainable.